<u>Retail Data Agent Task</u>

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Introduction & Objective

The objective of this task was to develop a functional Data Agent on Databricks using a retail sales dataset. I designed the agent's instructions, tested its performance with real business questions, and documented the entire process from design to execution.

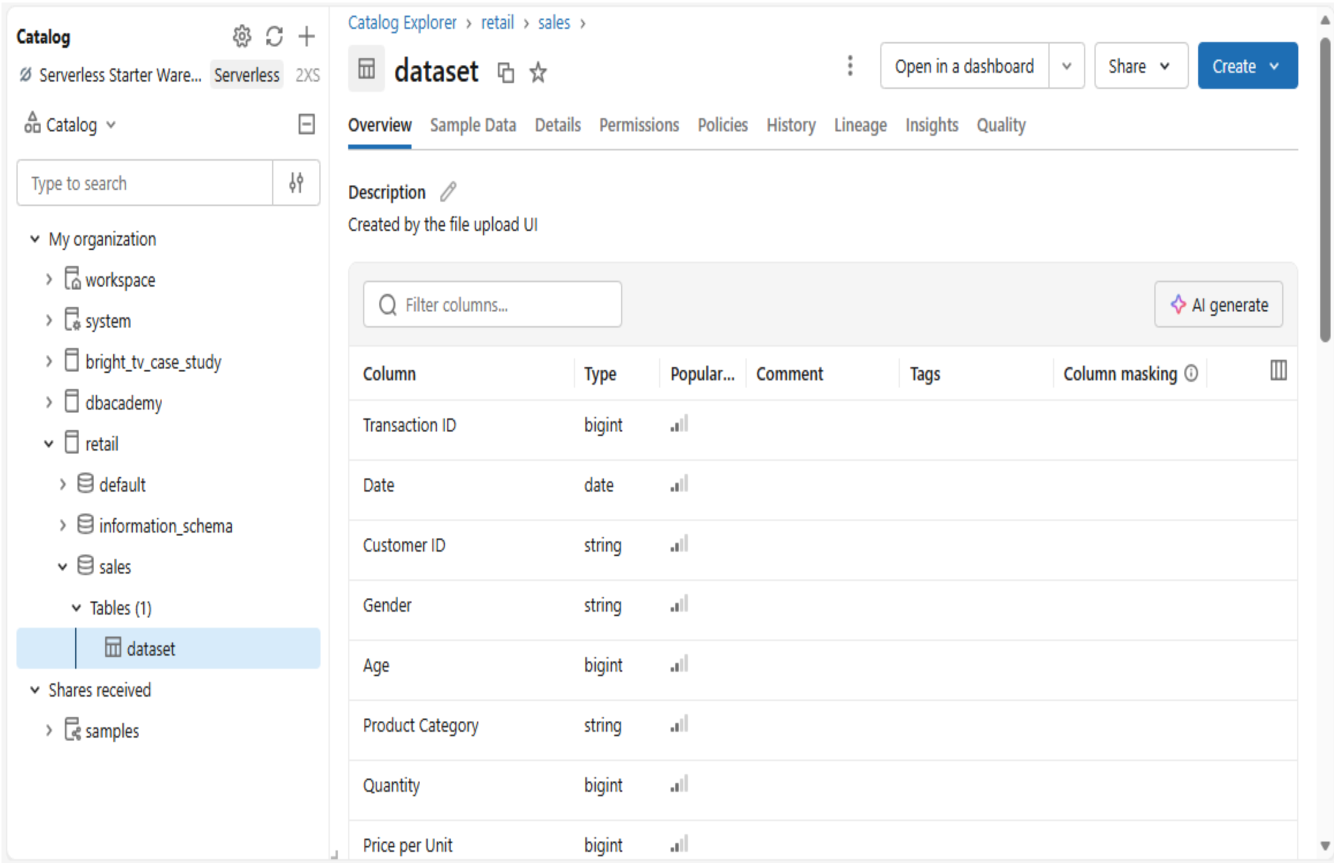
Dataset Description

The **retail_sales_data** dataset contains transactional records from a retail environment. Each row represents a single customer purchase, with details about the buyer, product, and sales value.

🗂️ **Columns Explained**

- **Transaction ID** → Unique identifier for each purchase.
- **Date** → The day the transaction occurred.
- **Customer ID** → Unique code for each customer (used for tracking, not personal details).
- **Gender** → Customer's gender (Male/Female).
- **Age** → Customer's age at the time of purchase.
- **Product Category** → Type of product purchased (Beauty, Clothing, Electronics).
- **Quantity** → Number of units bought in the transaction.
- **Price per Unit** → Cost of a single item in the category.

Steps 1–3: Set Up Your Data

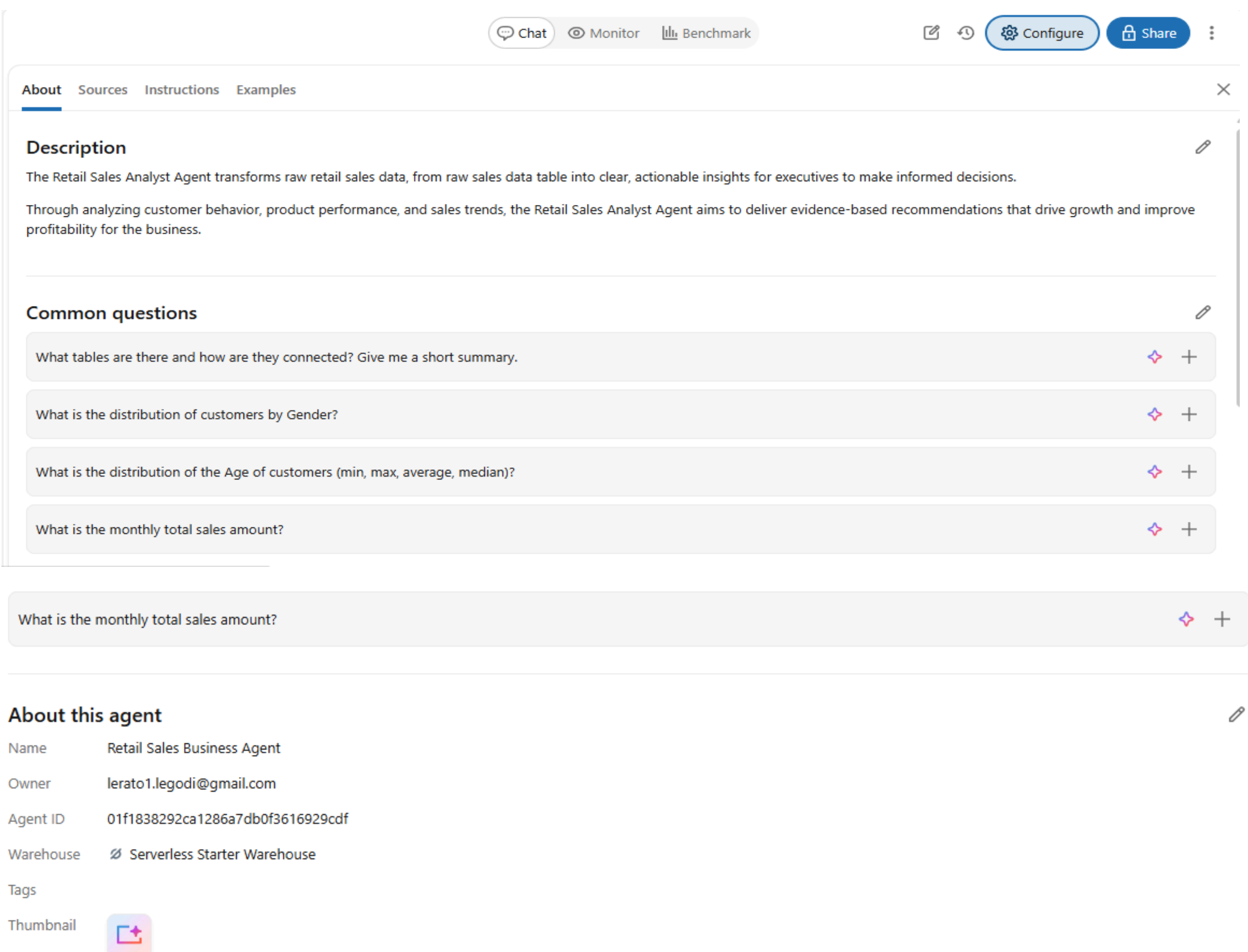


Step 4: Create the Data Agent

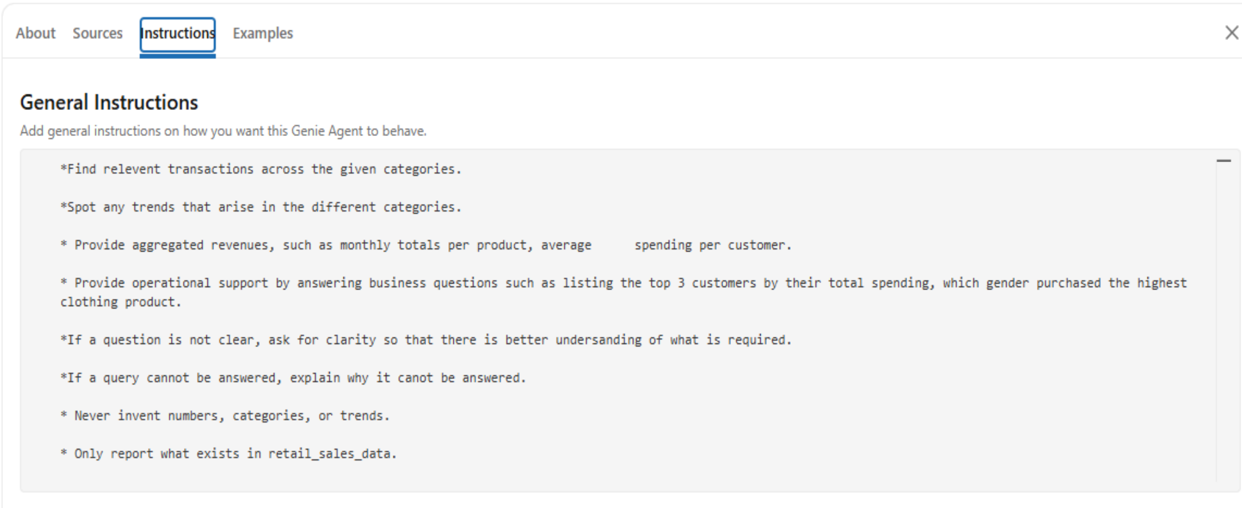
Retail Sales Business Agent

The Retail Sales Analyst Agent transforms raw retail sales data, from raw sales data table into clear, actionable insights for executives to make informed decisions.

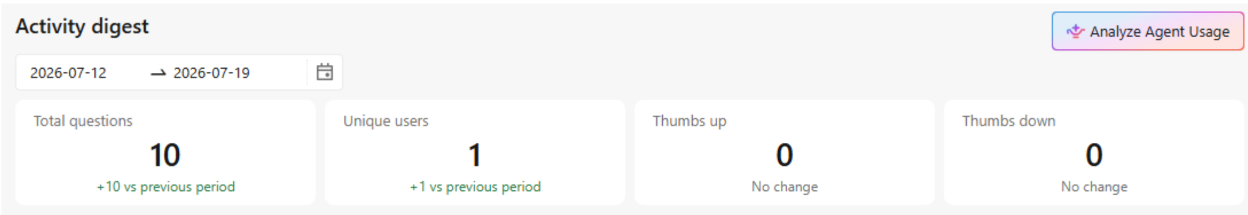
Through analyzing customer behavior, product performance, and sales trends, the Retail Sales Analyst Agent aims to deliver evidence-based recommendations that drive growth and improve profitability for the business.



Step 5: Write Your Own Instructions



Step 6: Test with 10 Questions



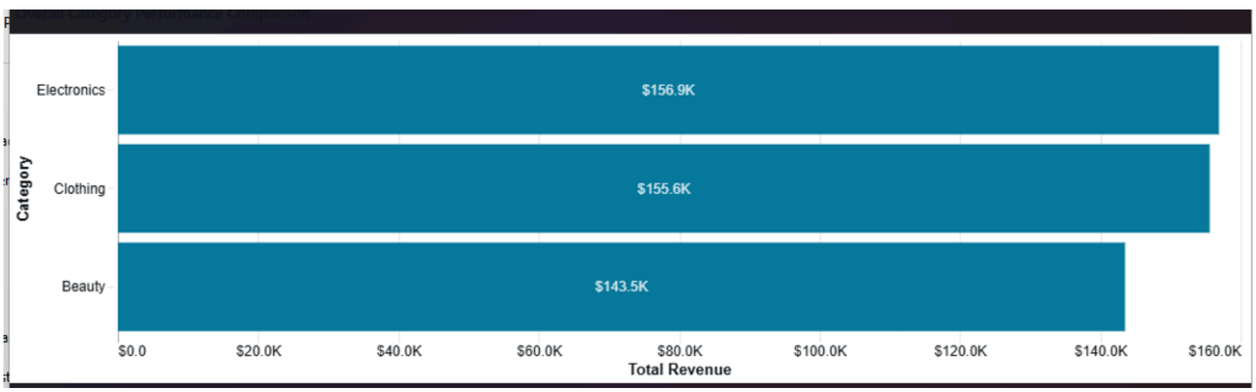
Spot any trends that arise in the different categories

Key Trends Across Product Categories

The data covers 1,000 transactions from January 1, 2023 to January 1, 2024, revealing distinct performance patterns across Beauty, Electronics, and Clothing categories. [1]

Overall Category Performance

Electronics leads in total revenue with $156,905, followed closely by Clothing at $155,580 and Beauty at $143,515. However, Beauty commands the **highest average transaction value** at $467.48, compared to Electronics ($458.79) and Clothing ($443.25). [2]



Which month in 2023 had the highest sales and for which product?

› Thinking complete

May 2023 had the highest sales in 2023 with total sales of **$53,150**, and **Electronics** was the top-selling product category that month with **$23,245** in sales. [1,2]

Month with highest sales in 2023

	month	total_sales
1	2023-05-01T00:00:00.000+00:...	53,150

Hide code

```sql
SELECT
  DATE_TRUNC('month', `Date`) as month,
  SUM(`Total Amount`) as total_sales
FROM
  `retail`.`sales`.`dataset`
WHERE
  `Date` >= '2023-01-01'
  AND `Date` < '2024-01-01'
GROUP BY
  DATE_TRUNC('month', `Date`)
ORDER BY
  total_sales DESC
LIMIT 1
```

Top product category in May 2023

	Product Category	total_sales
1	Electronics	23,245

Hide code

```
SELECT
  `Product Category`,
  SUM(`Total Amount`) as total_sales
FROM
  `retail`.`sales`.`dataset`
WHERE
  `Date` >= '2023-05-01'
  AND `Date` < '2023-06-01'
GROUP BY
  `Product Category`
ORDER BY
  total_sales DESC
LIMIT 1
```

Describe this dataset and the important KPIs/categories I can analyze

Retail Sales Dataset Overview

This dataset contains **1,000 retail transactions** from **1,000 unique customers** spanning **January 1, 2023 to January 1, 2024**, generating **$456,000 in total revenue.** [1]

Dataset Structure

The dataset includes transaction-level data with the following dimensions:

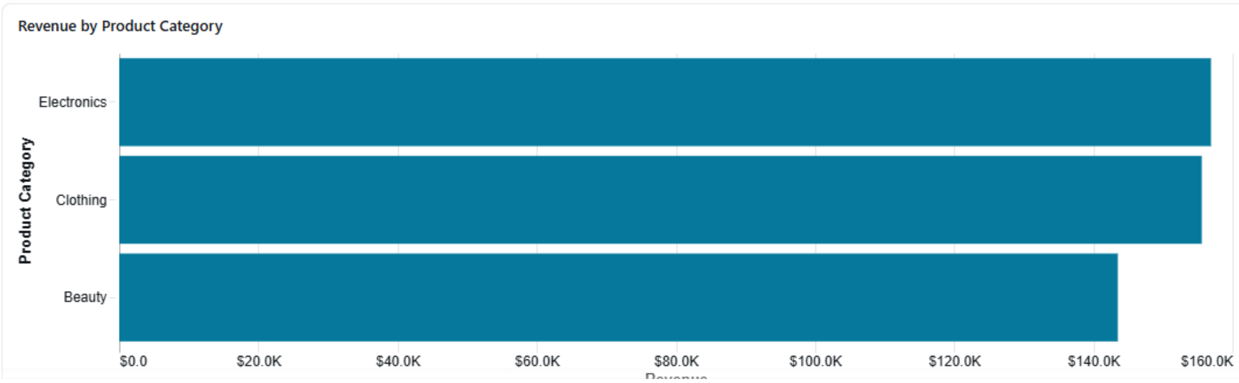
- **Customer attributes:** Customer ID, Gender, Age
- **Product information:** Product Category, Quantity, Price per Unit
- **Transaction details:** Transaction ID, Date, Total Amount

Key Demographics

Gender Distribution: The customer base is evenly split between Female (51%, 510 customers) and Male (49%, 490 customers), with nearly identical average transaction values of $456.55 and $455.43 respectively. [2]

Age Range: Customers range from **18 to 64 years old**, with an average age of 41 years and a median of 42 years. [3]

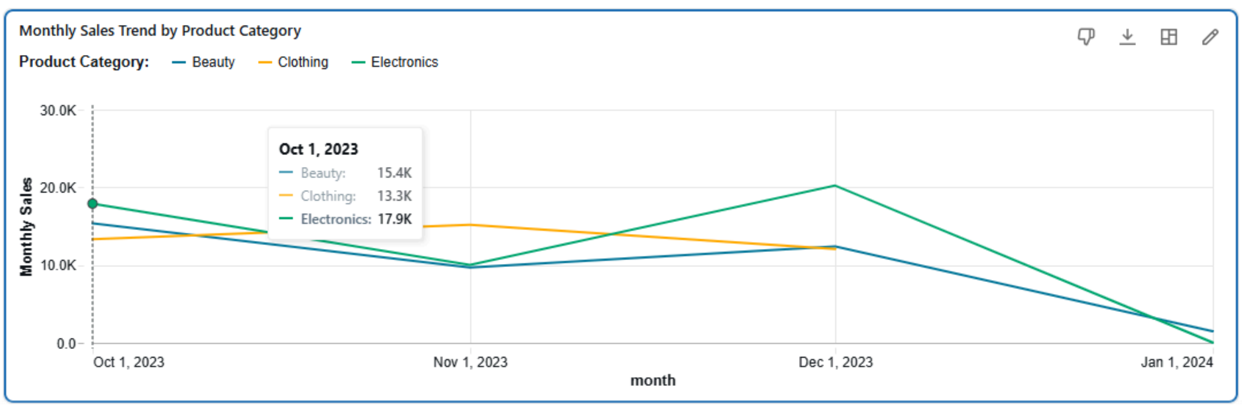
Product Categories & Performance



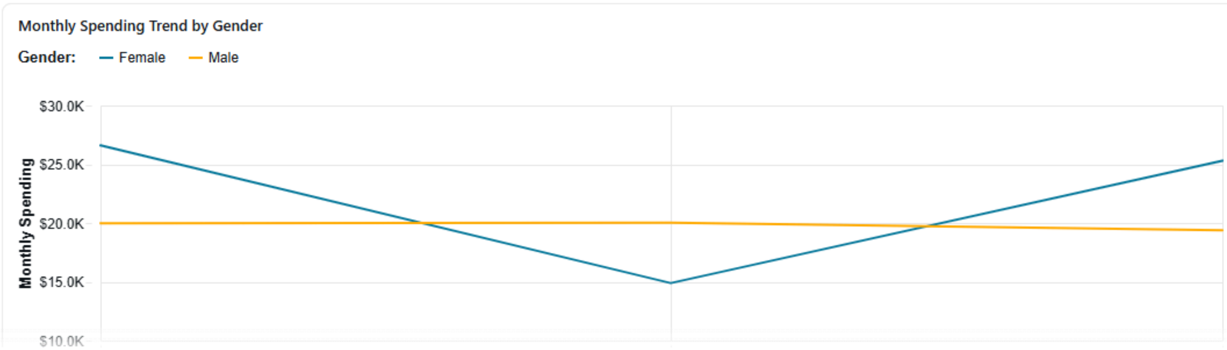
Sales by product category the last 3 months

Sales by Product Category - Last 3 Months

Electronics led sales with $48,180, followed by Clothing at $40,585 and Beauty at $38,955 over the October 2023 - January 2024 period. 1



which gender spent the most amount of money the last 3 months of the year



Category Preferences:

- **Female customers** spent most on Electronics (37.3% of their spending), followed by Clothing (35.1%) and Beauty (27.6%)
- **Male customers** also favored Electronics (39.1%), but allocated more to Beauty (32.0%) than Clothing (28.9%)
- Notably, male customers spent proportionally more on Beauty products compared to female customers [3]

provide trend that emerged in the beauty category the last 6 months

Beauty Category Trends (Last 6 Months)

The Beauty category shows **significant volatility** in the last six months (July 2023 - December 2023), with revenue ranging from $6.3K to $16.1K per month. [1]

Key Trends

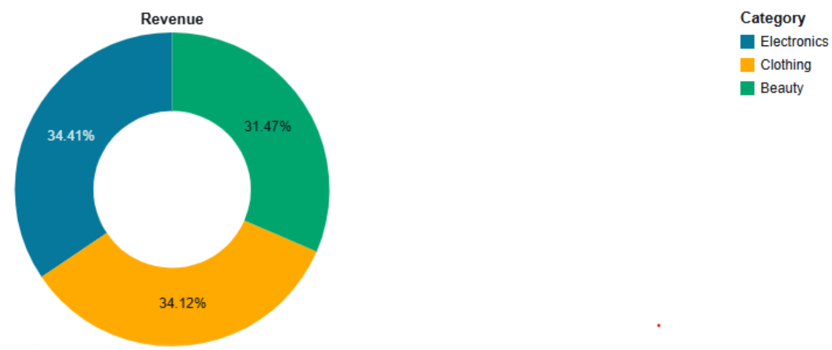
Revenue Volatility Pattern

Showcase the different visualization options with interesting aspects of the dataset

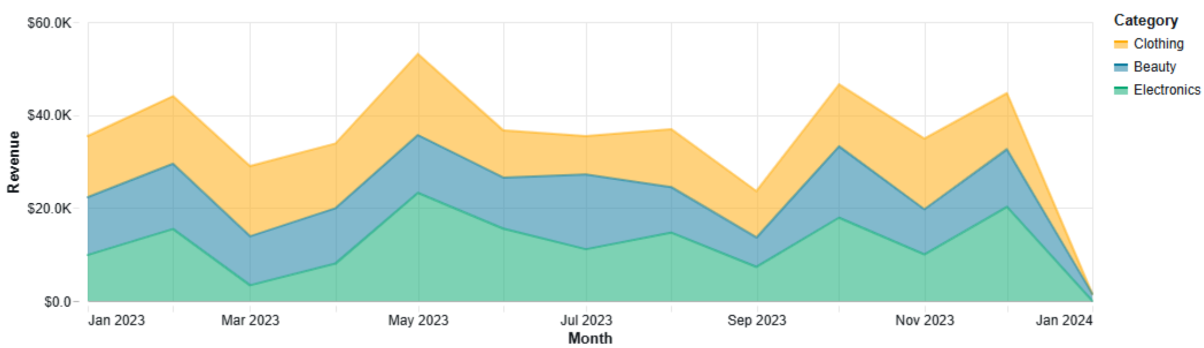
Retail Sales Dataset Visualization Showcase

This retail dataset spans **one year** (January 2023 - January 2024) with **1,000 transactions** from 1,000 unique customers across three product categories, generating **$456,000** in total revenue. [1]

Revenue Distribution by Product Category



Monthly Revenue Trends by Category



Category Performance Summary

Category	Total Revenue	Transactions
Electronics	$156,905	342
Clothing	$155,580	351
Beauty	$143,515	307

How many categories are there in the dataset?

There are 3 **product categories** in the dataset: Beauty, Electronics, and Clothing. 1

were clothes more popular that electronics with the male gender?

No, Electronics Were More Popular Than Clothing Among Male Customers

Among male customers, **Electronics generated higher revenue** ($80,170) compared to Clothing ($74,305), representing about 8% more revenue. 1

However, the popularity metrics show mixed signals:

- **Transaction count:** Clothing had slightly more transactions (177 vs 172)
- **Quantity sold:** Clothing had more units sold (453 vs 410)
- **Revenue:** Electronics generated more total revenue ($80,170 vs $74,305)

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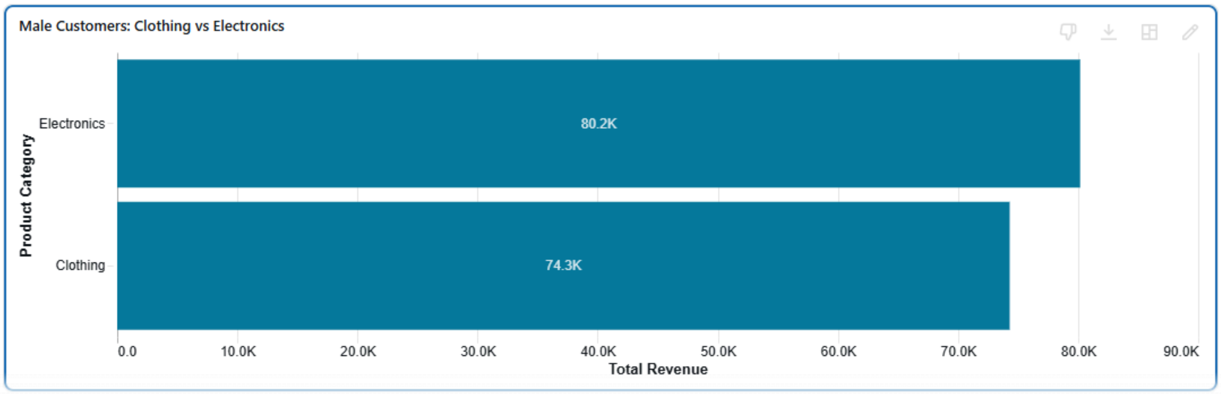

The higher revenue for Electronics despite fewer transactions suggests male customers spent more per purchase on Electronics, indicating stronger purchasing power and preference when measured by revenue rather than transaction frequency.

total purchases, revenue across all categories male vs female

Gender Comparison: Purchases and Revenue

Female customers lead with 510 purchases generating $232,840 in revenue, while male customers made 490 purchases totaling $223,160. The difference is modest—females account for 51% of purchases and 51% of revenue. 1

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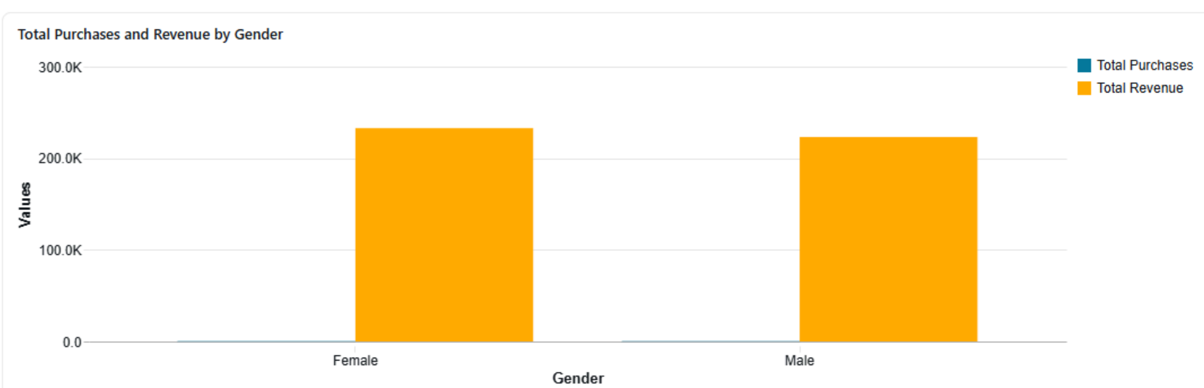
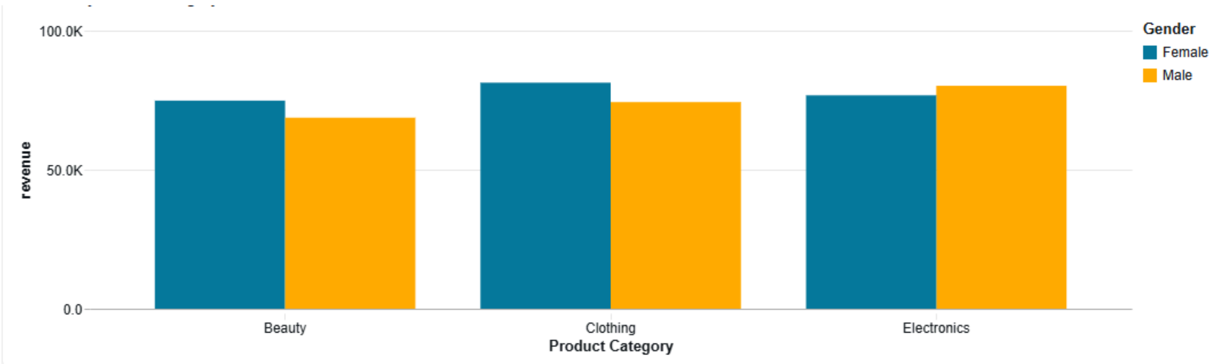


Category Breakdown

The gender split varies by product category [2]:

- **Beauty:** Female customers dominate with $74,830 revenue (166 purchases) vs. males at $68,685 (141 purchases)
- **Clothing:** Females lead in revenue at $81,275 (174 purchases) compared to males at $74,305 (177 purchases), despite males making slightly more purchases
- **Electronics:** Males edge ahead with $80,170 revenue (172 purchases) vs. females at $76,735 (170 purchases)



Source: Purchases and revenue by category and gender

Average transaction values are nearly identical: $456.55 for females and $455.43 for males. [1]